

BRANDEN RAYGOZA

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SUMMARY

Analytical aerospace engineering graduate with hands on experience in mechanical design, control systems, and hardware testing. Ability to conduct root cause investigations, perform risk assessments using fault tree analysis, and optimize product manufacturability. Eager to apply strong foundation in statics, thermodynamics, and hardware instrumentation to develop reliable, extreme environment mechanisms.

EDUCATION

California State Polytechnic University, Pomona — Bachelor of Science in Aerospace Engineering
Pomona, CA

May 2026
CPP GPA: 3.57 | Cumulative GPA: 3.30

- President's Honor List; Dean's Honor List
- Relevant coursework: Space Vehicle Dynamics, Controls, Thermodynamics, Aerospace Structures, Orbital Mechanics, Low Speed Aerodynamics, High Speed Aerodynamics, Avionics, Vibrations & Dynamics, CFD, Rocket Propulsion, Space Environment

MECHANICAL DESIGN & TESTING EXPERIENCE

Systems Deputy — Space Vehicle Design, Senior Design

Aug 2025 – May 2026

- Served as systems deputy, coordinating integration of subsystem designs and maintaining system-level requirements traceability
- Conducted risk analysis using fault tree methodology at system and component levels and determined mitigation strategies
- Derived first level requirements from mission objectives and decomposed into subsystem-level derived requirements
- Developed KDP milestone timeline based on analogous spacecraft missions and NASA program management handbooks
- Performed mission analysis to determine crew mass-collection capacity of mission duration
- Presented subsystem integration progress and risk analysis findings at formal design reviews

Hardware and Testing — S-Kubed, Small Satellite Stationkeeping

Aug 2025 – May 2026

- Contributed to hardware build, integration, and tests for a 3-axis Helmholtz cage simulating orbital geomagnetic environments
- Designed and 3D-printed a CubeSat mass model in SolidWorks and assembled printed components for hardware-in-the-loop testing
- Developed a MATLAB tool that determines placement of internal mass blocks to match a target moment-of-inertia tensor within 1% tolerance
- Authored formal test procedures and post-test reports documenting setup, methodology, and results
- Measure physical dimensions with vernier calipers to identify structural fasteners by specs to source cost-effective replacements

RELEVANT EXPERIENCE

Systems Liaison — Particle Image Velocimetry (PIV) Design Team

Jan 2024 – May 2025

- Operated PIV instrumentation; designed test-section components in SolidWorks including a delta wing for an existing wing mount
- Maintained team Gantt chart and coordinated cross-subsystem deliverables

Aero Feedback Control Systems Laboratory (ARO 3220L)

Aug 2024 – Dec 2025

- Modeled and deployed real-time PID controllers via Simulink to maneuver and stabilize dual-rotor Quanser Aero System
- Conducted hardware-in-the-loop testing to analyze dynamic response and tune controllers for optimal system performance

Wind Tunnel Testing Laboratory (ARO 4351L)

Jan 2024 – May 2024

- Operated low-speed and high-speed wind tunnel and performed post-run data analysis; replaced high-speed wind tunnel model

Aerospace Structures Laboratory (ARO 3570L)

Aug 2023 – Dec 2023

- Performed structural testing on a Shimadzu AGS-X universal test machine: tension, compression, three-point bending, cylinder buckling

Lead Proposal Review Panelist — NASA L'SPACE Academy

May 2022 – Aug 2022

- Selected as lead reviewer for peer mission proposals; authored a Phase-1-selected proposal for an aviation vibration-damping structure

EMPLOYMENT

Math Tutor — East Los Angeles College Math Lab

August 2022 – Present

- Tutor college-level mathematics across disciplines and skill levels, adapting instruction to diverse learning needs

Stocker / Cashier / Acting Asst. Manager — Daiso California LLC

June 2017 – August 2022

- Trained new hires, conducted interviews, and managed register operations with 100% accuracy

ADDITIONAL HANDS-ON EXPERIENCE

- Disassemble, diagnose, and repair consumer electronics at the component level; solder and maintain small-electronics hardware
- Independently perform automotive and motorcycle maintenance and repairs using owned tools, multimeter, and service documentation

SKILLS

CAD & Drafting: SolidWorks, AutoCAD, hand drafting | **Software:** MATLAB, Simulink, Excel, Git, MS Planner

Test & Fab: Multimeter, soldering, hardware-in-the-loop testing, PID control, 3D printing, hand tools | **Languages:** English, Spanish